

SKILL 15

```
nishanth@LAPTOP-6Q3ON07H:~$ mkdir task15
nishanth@LAPTOP-6Q3ON07H:~$ cd task15
nishanth@LAPTOP-6Q3ON07H:~/task15$ nano task15.c
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

int main() {
    int fd;

    printf("=== Combined Redirection Test ===\n");

    fd = open("combined.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("open");
        return 1;
    }
}
```

```
int saved_stdout = dup(STDOUT_FILENO);
int saved_stderr = dup(STDERR_FILENO);

dup2(fd, STDOUT_FILENO);
dup2(fd, STDERR_FILENO);

printf("This is standard output\n");
fprintf(stderr, "This is standard error\n");

close(fd);

fflush(stdout);
fflush(stderr);

dup2(saved_stdout, STDOUT_FILENO);
dup2(saved_stderr, STDERR_FILENO);

close(saved_stdout);
close(saved_stderr);

printf("stdout and stderr merged into combined.txt\n");

printf("\n=== Verify Combined Output ===\n");

FILE *fp = fopen("combined.txt", "r");
char line[200];

if (fp != NULL) {
    while (fgets(line, sizeof(line), fp) != NULL) {
        printf("%s", line);
    }
    fclose(fp);
}
```

```
printf("\n=== Complex Command Test ===\n");
printf("Command: ls -l | grep .c > result.txt\n");

int pipefd[2];

if (pipe(pipefd) == -1) {
    perror("pipe");
    return 1;
}

pid_t pid1 = fork();

if (pid1 == 0) {
    dup2(pipefd[1], STDOUT_FILENO);

    close(pipefd[0]);
    close(pipefd[1]);

    execlp("ls", "ls", "-l", NULL);

    perror("ls failed");
    exit(1);
}
```

```
pid_t pid2 = fork();

if (pid2 == 0) {
    fd = open("result.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("result.txt");
        exit(1);
    }

    dup2(pipefd[0], STDIN_FILENO);
    dup2(fd, STDOUT_FILENO);

    close(fd);
    close(pipefd[0]);
    close(pipefd[1]);

    execlp("grep", "grep", ".c", NULL);

    perror("grep failed");
    exit(1);
}
```

```
dup2(pipefd[0], STDIN_FILENO);
dup2(fd, STDOUT_FILENO);

close(fd);
close(pipefd[0]);
close(pipefd[1]);

execlp("grep", "grep", ".c", NULL);

perror("grep failed");
exit(1);
}

close(pipefd[0]);
close(pipefd[1]);

waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);
```

```
printf("Pipeline and redirection executed successfully\n");

printf("\n=== Execution Plan ===\n");
printf("Step 1: Execute ls -l\n");
printf("Step 2: Send output through pipe\n");
printf("Step 3: grep filters .c files\n");
printf("Step 4: Redirect output to result.txt\n");

printf("\n=== Syntax Validation Test ===\n");

char *valid = "ls -l | grep .c > result.txt";

if (valid != NULL)
    printf("Syntax accepted: %s\n", valid);
else
    printf("Invalid syntax\n");

printf("\n=== Edge Case Test ===\n");

if (system("cat missing_file.txt > /dev/null 2>&1") != 0) {
    printf("Missing file error handled successfully\n");
}

printf("\n=== Verify result.txt ===\n");
```

```
fp = fopen("result.txt", "r");

if (fp != NULL) {
    while (fgets(line, sizeof(line), fp) != NULL) {
        printf("%s", line);
    }

    fclose(fp);
}

printf("\n=== Task 15 Complete ===\n");

return 0;
}
```

```
=== Combined Redirection Test ===
stdout and stderr merged into combined.txt

=== Verify Combined Output ===
This is standard output
This is standard error

=== Complex Command Test ===
Command: ls -l | grep .c > result.txt
Pipeline and redirection executed successfully
```

```
=== Execution Plan ===
Step 1: Execute ls -l
Step 2: Send output through pipe
Step 3: grep filters .c files
Step 4: Redirect output to result.txt

=== Syntax Validation Test ===
Syntax accepted: ls -l | grep .c > result.txt

=== Edge Case Test ===
Missing file error handled successfully
```

```
=== Verify result.txt ===  
-rw-r--r-- 1 nishanth nishanth  47 Oct  1 16:29 combined.txt  
-rw-r--r-- 1 nishanth nishanth   0 Oct  1 16:29 result.txt  
-rwxr-xr-x 1 nishanth nishanth 16768 Oct  1 16:29 task15  
-rw-r--r-- 1 nishanth nishanth  3035 Oct  1 16:29 task15.c
```

Then verify both generated files:

cat combined.txt

cat result.txt

```
nishanth@LAPTOP-6Q3ON07H:~/task15$ cat combined.txt  
This is standard output  
This is standard error  
nishanth@LAPTOP-6Q3ON07H:~/task15$ cat result.txt  
-rw-r--r-- 1 nishanth nishanth  47 Oct  1 16:29 combined.txt  
-rw-r--r-- 1 nishanth nishanth   0 Oct  1 16:29 result.txt  
-rwxr-xr-x 1 nishanth nishanth 16768 Oct  1 16:29 task15  
-rw-r--r-- 1 nishanth nishanth  3035 Oct  1 16:29 task15.c
```